

Advanced Geovisualisation

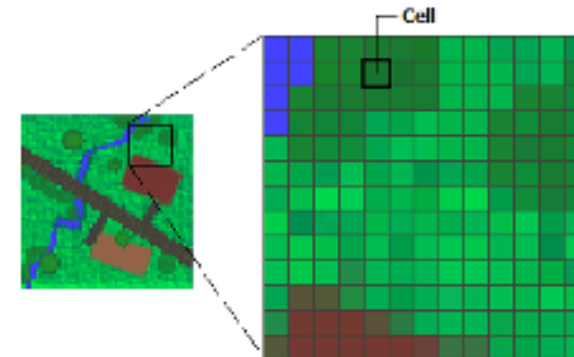
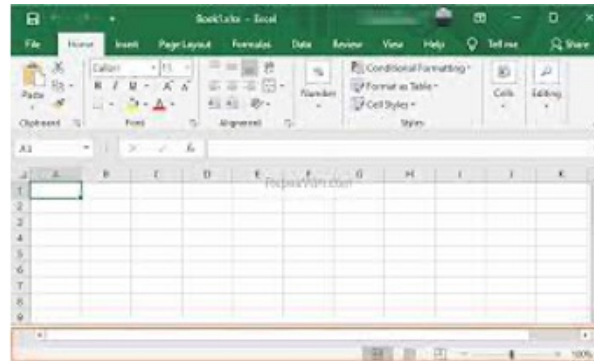
From Pixels to Patterns

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What is a Raster?

- A **grid of pixels** — each cell holds a value
- Divides 2D space into **regular, equal-sized cells**
- Pixel values can represent:
 - **Continuous** variables — elevation, temperature, light intensity
 - **Categorical** variables — land cover class, soil type
- Value assigned by mean, centre point, or aggregation rule

File Formats



Types of Raster Data

Grayscale Rasters

One band, continuous values

- The simplest raster: a **single band** where pixel brightness encodes magnitude
- Examples: nightlights intensity, radar backscatter, thermal emission



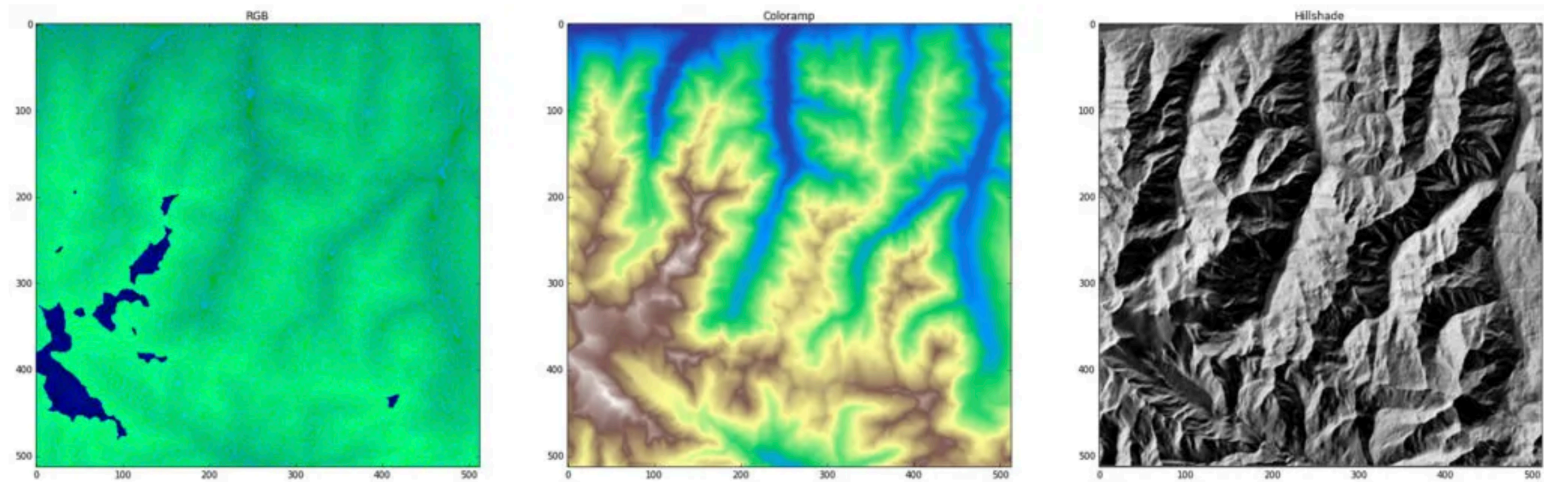
Nightlights Data

Human activity, visible from space

- Darker areas → lower artificial light; brighter areas → higher intensity
- Pixel values represent **radiance or luminance** of nighttime lights
- Applications:
 - Monitoring **urban growth and sprawl**
 - Assessing **light pollution**
 - Proxying **economic activity** in data-sparse regions

Elevation Rasters

The shape of the land



Left: raw RGB — Centre: hypsometric tint — Right: hillshade

Mapbox

Digital Elevation Models (DEMs)

- Each pixel value = **elevation above sea level** at that location
- Foundation for terrain analysis, hydrological modelling, flood risk
- Derived products: slope, aspect, curvature, viewshed
- Widely used in infrastructure planning, disaster response, and ecology

Multispectral Imagery

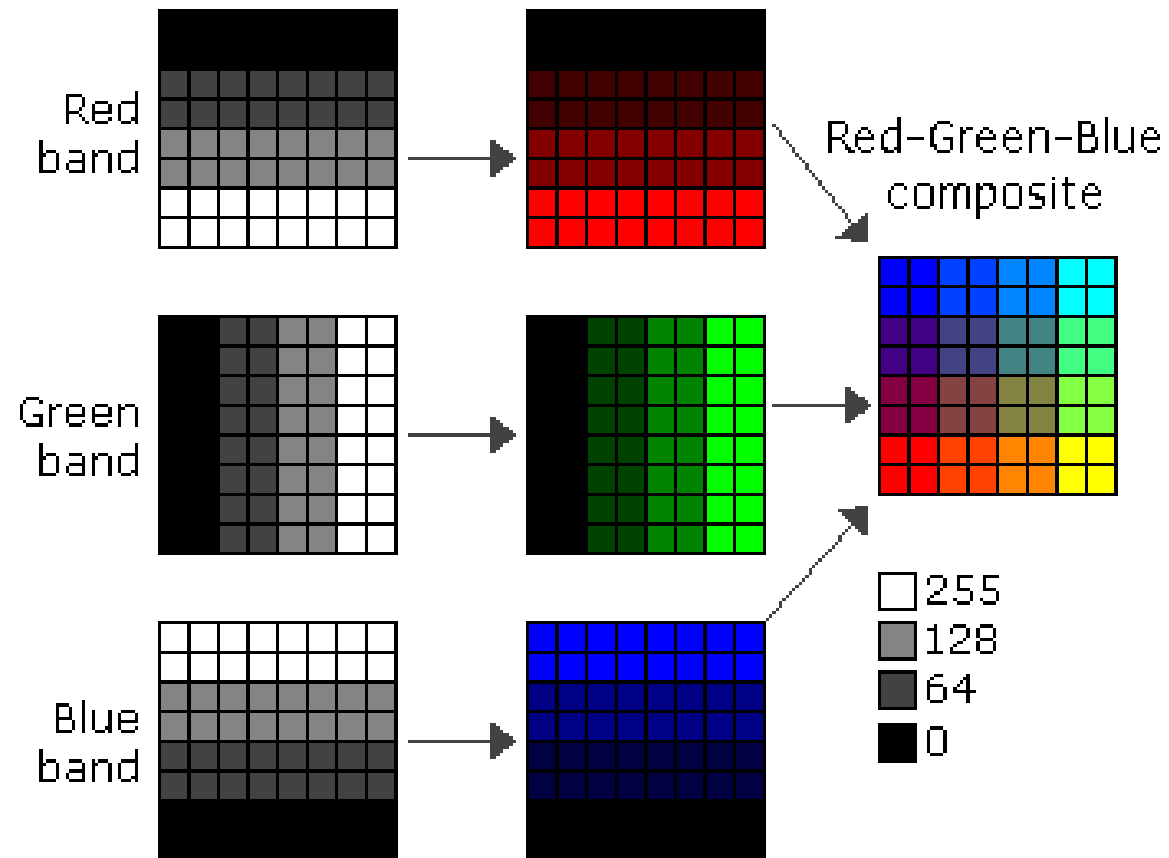
Beyond Visible Light

What sensors can see that we can't

- Human eyes perceive **3 bands** (red, green, blue)
- Multispectral sensors capture **4–12+ bands** across the electromagnetic spectrum
- Each additional band reveals something **invisible to the naked eye**:
 - Plant health, soil moisture, surface temperature, mineral composition
- Different surfaces have distinct **spectral signatures** — fingerprints in light

The landscape looks very different depending on which wavelengths you're listening to.

Multispectral Rasters

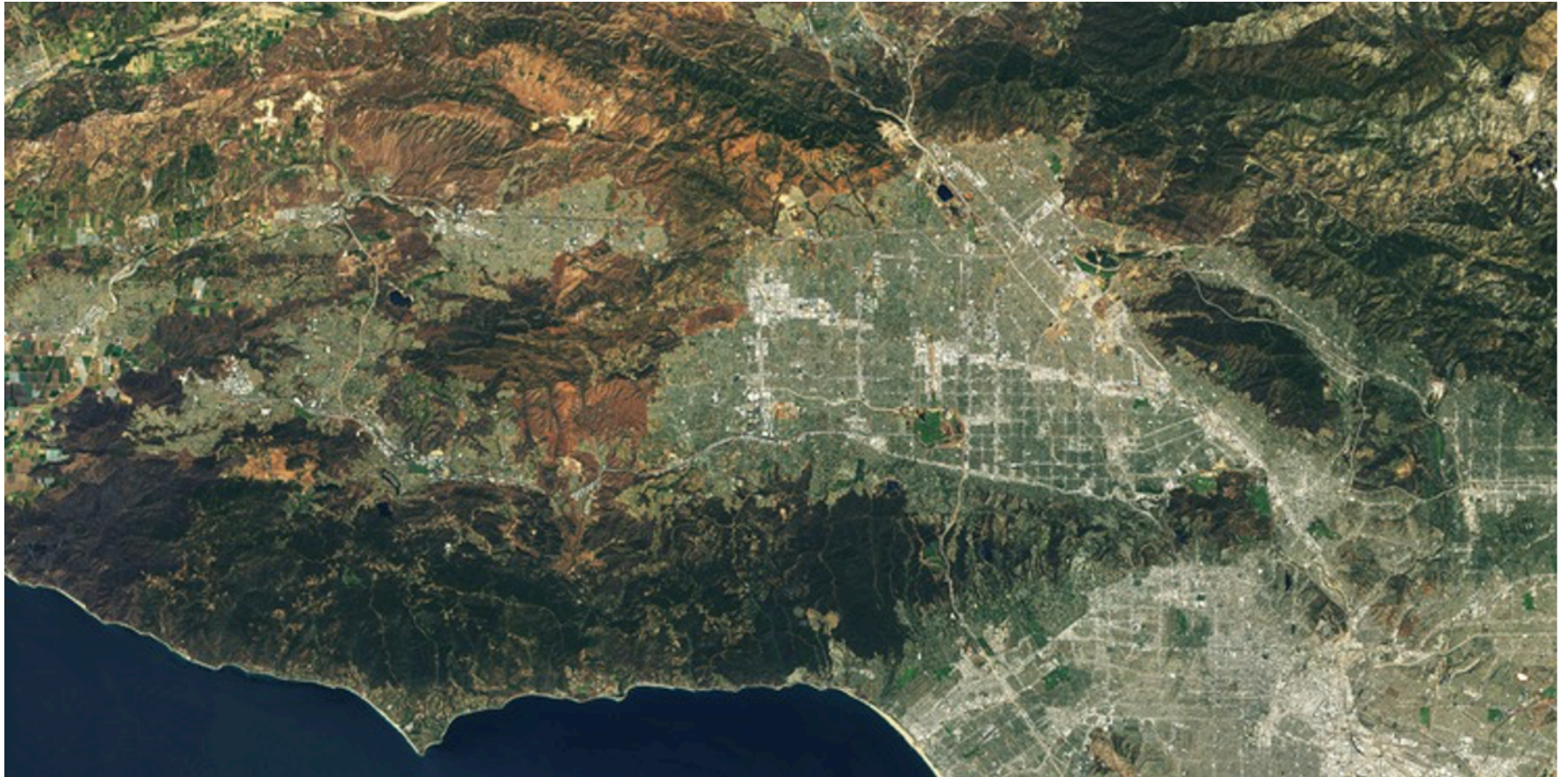


Landsat Band Table

Band	Name	Wavelength	Resolution	What it's good for
1	Ultra Blue (Coastal/Aerosol)	0.43–0.45 μm	30 m	Coastal water, aerosol studies
2	Blue	0.45–0.51 μm	30 m	Water depth, soil/vegetation distinction
3	Green	0.53–0.59 μm	30 m	Peak vegetation reflectance
4	Red	0.64–0.67 μm	30 m	Chlorophyll absorption; vegetation vs bare soil
5	Near Infrared (NIR)	0.85–0.88 μm	30 m	Biomass, vegetation health, water body edges
6	SWIR-1	1.57–1.65 μm	30 m	Soil moisture, snow/cloud differentiation
7	SWIR-2	2.11–2.29 μm	30 m	Geology, mineralogy, fire burn scars
8	Panchromatic	0.50–0.68 μm	15 m	High-resolution greyscale — pan-sharpening
9	Cirrus	1.36–1.38 μm	30 m	Detecting high-altitude cirrus cloud contamination
10	Thermal Infrared (TIRS-1)	10.6–11.19 μm	100 m	Surface temperature
11	Thermal Infrared (TIRS-2)	11.50–12.51 μm	100 m	Surface temperature (more accurate dual-band retrieval)

True Colour: Bands 2, 3 and 4

Blue, green, red — what the eye would see



Resolution: The Fundamental Trade-off

Spatial Resolution

Can you see what you're looking for?

- Defines the **smallest distinguishable feature** in an image
- A 1m pixel $\neq$ a 10m pixel — the difference is *a building vs. a blur*
- **Higher resolution** $\rightarrow$ **richer insight** into land use, urban change, disaster damage
- But there's a cost: **larger files, more compute, harder to scale**

Temporal Resolution

When did that change — and how fast?

- How **frequently** a sensor revisits the same location
- High temporal resolution lets us watch the world **unfold in near real-time**:
 - Flood progression
 - Crop growth cycles
 - Urban expansion
- The faster things change, the more temporal resolution matters

Data Volume & Storage

The deluge is already here

- Modern satellites generate **terabytes per day** — and it's accelerating
- Storage isn't just a technical problem, it's a **scientific bottleneck**
- Traditional on-premise hardware: expensive, slow to scale
- Cloud-based storage: flexible, but raises questions of **cost, access, and sovereignty**

Where to Get Imagery

- **High spatial resolution** — costly commercial products
- **Lower spatial resolution** — free via NASA, ESA, and others:
 - [USGS Earth Explorer](#) — Landsat, MODIS, and more
 - [NASA Earthdata Search](#) — full NASA Earth observation catalogue
 - [Copernicus Open Access Hub](#) — ESA Sentinel imagery (1, 2, 3, 5P)
 - [AppEEARS \(LP DAAC\)](#) — easy subsetting of NASA land products
 - [Landsat on AWS](#) — cloud-hosted Landsat archive

Supervised Classification

Turning Pixels into Meaning

What are we actually doing?

- Every pixel in a multispectral image has a **spectral signature** — a fingerprint of how it reflects light across bands
- Supervised classification asks: **can we teach an algorithm to recognise these fingerprints?**
- Goal: assign every pixel a land cover label — urban, rural, water, vegetation...

Why Multispectral for Classification?

More bands = more separability

- Standard RGB: **3 bands** — multispectral sensors: **4–12+**
- Key bands for urban/rural distinction:
 - **NIR** → healthy vegetation glows; concrete doesn't
 - **Red Edge** → separates crops from forests
 - **SWIR** → distinguishes built surfaces and bare soil
- More spectral dimensions = classes that overlap in RGB **pull apart**

Training Points

The human input the algorithm depends on

- You must provide **labelled examples** of each class — this is the “supervised” part
- Training points tell the classifier: *“this is what urban looks like spectrally”*
- Quality requirements:
 - **Representative** — capture the variety within each class
 - **Balanced** — avoid over-representing one class
 - **Accurate** — wrong labels poison the model
- Collected via: field survey, manual digitising, or existing land cover datasets

The Classification Pipeline

From labelled points to labelled map

1. **Define classes** — urban, peri-urban, agriculture, forest, water...
2. **Collect training points** — ideally 50–200+ samples per class
3. **Extract spectral values** at training locations
4. **Train a classifier** — Random Forest, SVM, Maximum Likelihood...
5. **Apply to full image** — every pixel gets a label
6. **Validate** — withhold test points, compute accuracy metrics

The pipeline is straightforward. The judgement calls within it are not.

Accuracy Assessment

How do we know if it worked?

- Withhold a **test set** (20–30% of labelled points) — never used in training
- Build a **confusion matrix**: where does the classifier get confused?
 - Urban misclassified as bare soil?
 - Dense vegetation confused with rural settlements?
- Key metrics: **Overall Accuracy, Kappa coefficient, Per-class F1**
- Urban/rural boundaries are notoriously **fuzzy** — peri-urban is hard

A 90% accurate map still means 1 in 10 pixels is wrong — at scale, that matters.



The Cloud Problem

- Clouds are **opaque to optical sensors** — where there's cloud, there's no data
- In tropical regions, cloud cover can exceed **80% of observations** year-round
- A single cloudy acquisition can ruin a time-sensitive classification
- Cloud **shadows** are almost as problematic — spectrally dark, easily confused with water or urban areas

Questions



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